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Numerical and Experimental Studies of the Attenuation Characteristics of Friction Torque in a Wet Multidisc Clutch

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Abstract: A numerical model is proposed to calculate both the contact pressure and the friction torque (FT) of the friction components considering the spline friction in a wet multidisc clutch, which is verified by the bench test. The results indicate that the spline friction of components is an important factor causing the axial attenuation of contact pressure on friction pairs. As the applied pressure increases, the attenuation amplitude of contact pressure increases gradually. In addition, the average single-pair FT decreases with the increasing number of friction pairs, thus leading to the decrease of the growth rate of total FT. Therefore, when the number of friction pairs reaches a certain number, it is not reliable to obtain a good torque enhancement, indicating that the effect of spline friction needs to be weakened to reduce the attenuation of contact pressure.

Keywords: wet multidisc clutch; friction torque; contact pressure; spline friction

Citation: Chen, M.; Zhang, B.; Feng, Y.; Wang, L.; Wang, H. Numerical and Experimental Studies of the Attenuation Characteristics of Friction Torque in a Wet Multidisc Clutch. *Appl. Sci.* **2021**, *11*, 814. https://doi.org/10.3390/app11020814

Received: 15 December 2020
Accepted: 13 January 2021
Published: 16 January 2021

1. Introduction

The wet multidisc clutch plays an important role in power-shift transmission, determining stability and reliability during the process of starting and shifting [1,2]. The engagement properties of friction components have an immediate impact on the torque capacity of wet clutch [3–5]. With the aim to improve the torque efficiency and service life of the wet clutch, numerous theoretical and experimental researches were conducted regarding the thermodynamic characteristics. Based on the transient temperature model, Cui et al. [6] and Xie et al. [7] revealed that the grooved regions of the friction component could reduce the temperature gradient and deformation effectively. Marklund et al. [8] proposed a torque transmission model to predict the friction torque (FT) and surface temperature of the friction pair. In addition, an algorithm to estimate the FT and lifetime of the clutch was derived with the application of Reye's theory and Archard's law [9].

As shown in Figure 1, the spline teeth of the friction components have obvious wear marks after the experiments, indicating that the spline teeth were subjected to friction during the axial movement of friction components. However, relatively limited studies were devoted to the comprehensive analysis of the clutch structure and taking the spline friction into account. The paper [10] offers a detailed mathematical analysis, given the spline friction, to investigate the various consequences on the FT of clutch. Newcomb et al. [11] developed a contact pressure model of the friction pair to analyze the interface temperature distribution in view of the spline friction. Moreover, by using the thermal model of multidisc clutch, Yu et al. [12] investigated the influence of spline friction on the temperature difference between different friction pairs.

应用科学

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文章湿式多片离合器摩擦力矩衰减特性的数值与实验研究

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摘要：提出了一种数值模型，用于计算考虑花键摩擦的湿式多片离合器中摩擦元件的接触压力和摩擦力矩（FT），并通过台架试验验证。结果表明，元件的花键摩擦是导致摩擦副接触压力轴向衰减的重要因素。随着施加压力的增加，接触压力的衰减幅度逐渐增大。此外，平均单对摩擦扭矩随摩擦副数量的增加而减小，从而导致总摩擦力矩增长率的下降。因此，当摩擦副数量达到一定值时，通过增加摩擦副数量来获得良好的扭矩增强并不可靠，表明需要削弱花键摩擦的影响以减少接触压力的衰减。

关键词：湿式多片离合器；摩擦力矩；接触压力；花键摩擦

引用：Chen, M.; Zhang, B.; Feng, Y.; Wang, L.; Wang, H. 湿式多片离合器摩擦力矩衰减特性的数值与实验研究.应用科学**2021**,*11*, 814.
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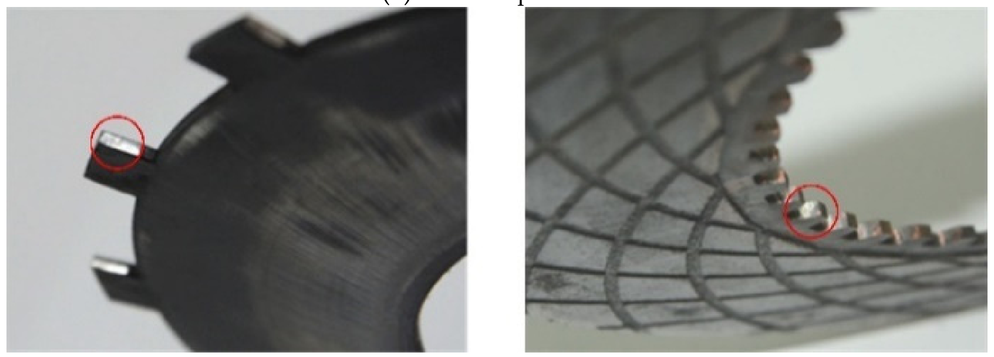
接收日期：2020年12月15日
接受日期：2021年1月13日
出版日期：2021年1月16日

1. 引言

湿式多片离合器在动力换挡变速器中起着重要作用，决定了启动和换挡过程中的稳定性和可靠性 [1,2]。摩擦元件的接合特性直接影响湿式离合器的扭矩容量 [3–5]。为了提高湿式离合器的扭矩效率和使用寿命，进行了大量关于热力学特性的理论和实验研究。基于瞬态温度模型，Cui 等人 [6]和Xie 等人 [7] 揭示了摩擦元件的沟槽区域能有效减小温度梯度和变形。Marklund 等人 [8] 提出了一种扭矩传递模型来预测摩擦副的摩擦力矩（FT）和表面温度。此外，基于Reye理论和Archard定律推导了一种估算离合器FT和寿命的算法 [9]。

如图1所示，摩擦元件的花键齿在实验后有明显磨损痕迹，表明花键齿在摩擦组件轴向移动过程中受到摩擦。然而，针对离合器结构综合考虑花键摩擦的详细研究相对有限。本文 [10] 提供了详细的数学分析，考虑花键摩擦，研究其对离合器FT的各种影响。Newcomb 等人 [11]建立了摩擦副的接触压力模型，以分析考虑花键摩擦的界面温度分布。此外，通过采用多盘离合器的热模型，Yu 等人 [12]研究了花键摩擦对不同摩擦副之间温差的影响。

(**a**) Before experiment



(**b**) After experiment

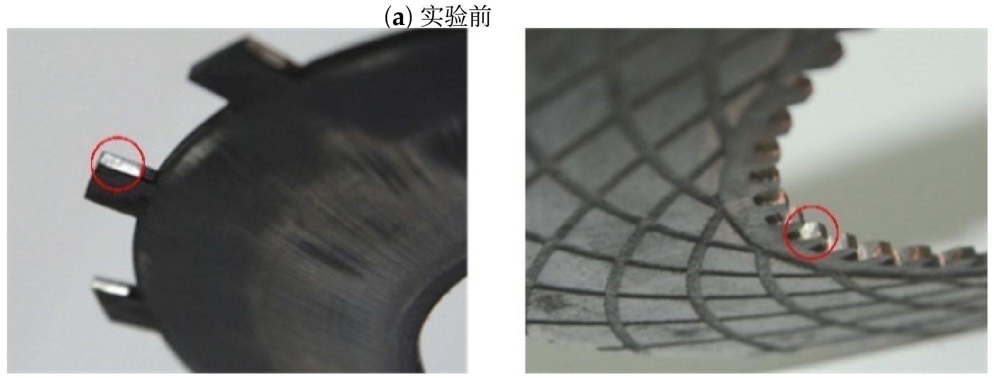
Figure 1. Wear of the spline teeth.

This paper is devoted to investigating the effect of spline friction on the contact pressure and FT of friction components. The mathematical model regarding the spline friction of components is established first. Then, the numerical simulation and bench test are carried out by changing the number of friction pairs to verify the existence of spline friction. Finally, the effect of spline friction is summarized in detail, and the measures to improve the torque capacity and lifetime of multidisc clutch are proposed.

2. Numerical Model

Figure 2 presents a three-dimensional exploded layout of the wet clutch visually. The separate plates and friction discs are arranged alternately, and respectively mounted to the cylinder liner and driving shaft by splines [13]. The circlip installed in the groove of cylinder liner plays a role in limiting the axial position of friction components [14]. By pumping pressure oil into the piston chamber, the friction components start to rub against each other, and the axial gap between them gradually disappears [15,16]. Meanwhile, the separate plates, as the driven part, transfer the input power to the transmission system. Finally, the dynamic friction of spline teeth converts to static friction after the gap is eliminated [17].

(**a**) 实验前



(**b**) 实验后

图1. 花键齿的磨损。

本文致力于研究花键摩擦对摩擦元件接触压力和摩擦力矩的影响。首先建立了关于组件花键摩擦的数学模型。然后，通过改变摩擦副数量进行数值模拟和台架试验，验证花键摩擦的存在。最后，详细总结了花键摩擦的影响，并提出了提高多盘离合器扭矩容量和寿命的措施。

2. 数值模型

图2 以三维爆炸图的形式直观展示了湿式离合器。分离片和摩擦盘交替排列，分别通过花键 [13]安装在缸套和驱动轴上。安装在缸套槽中的卡簧起到限制摩擦组件轴向位置的作用 [14]。通过将压力油泵入活塞腔，摩擦组件开始相互摩擦，它们之间的轴向间隙逐渐消失 [15,16]。同时，作为从动件的分离片将输入功率传递至传动系统。最后，间隙消除后，花键齿的动摩擦转变为静摩擦 [17]。

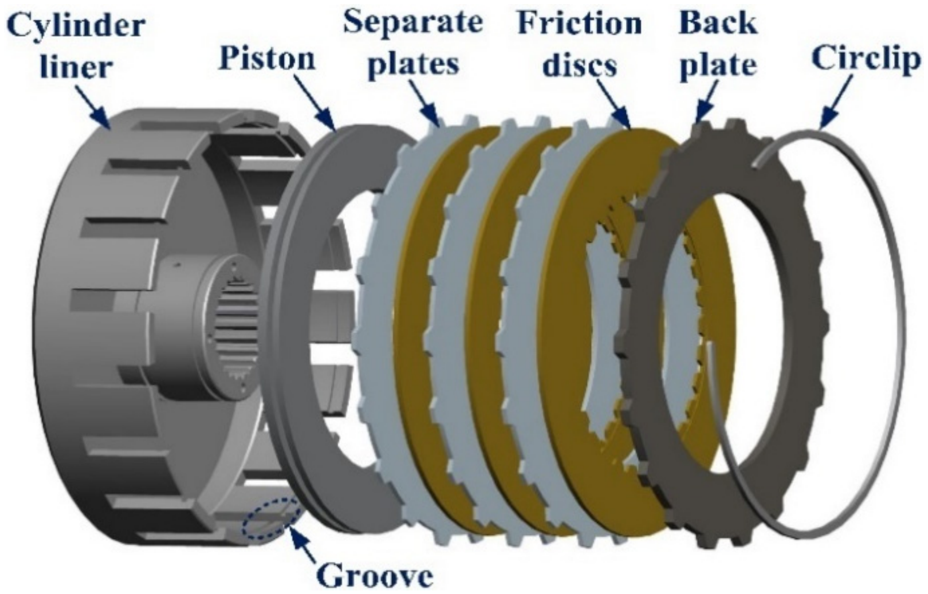


Figure 2. Exploded layout of the multidisc clutch.

2.1. Spline Friction

Since the structural characteristics of the friction disc and separate plate are similar, the separate plate is chosen for the theoretical analysis and formula derivation. As shown in Figure 3, during the constant speed sliding process of the clutch, the separate plate reaches equilibrium status under the combined action of FT T_f and cylinder liner resistance torque T_s. The friction force, nominal tangential force and normal force acting on the spline teeth are expressed as F_s, F_{cs} and F_{ns}, respectively. Moreover, F_p is the axial contact pressure applied on the separate plate. Hence, the equilibrium equation can be written as

$$-T_s = T_f = -N_s F_{cs} R_s \tag{1}$$

where R_s is the pitch radius, N_s is the total number of splines in a single separate plate. It should be noted that the clockwise direction is positive, and the subscripts *s* and *f* represent separate plate and friction disc, respectively.

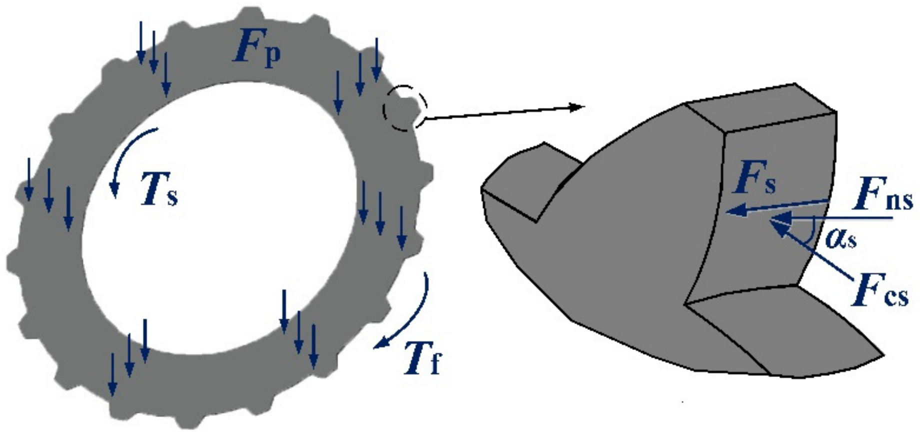


Figure 3. Force diagram of the separate plate.

The normal force that cylinder liner acts on the involute spline can be given as

$$F_{ns} = \frac{F_{cs}}{\cos \alpha_s} \tag{2}$$

where α_s is the pressure angle of spline teeth.

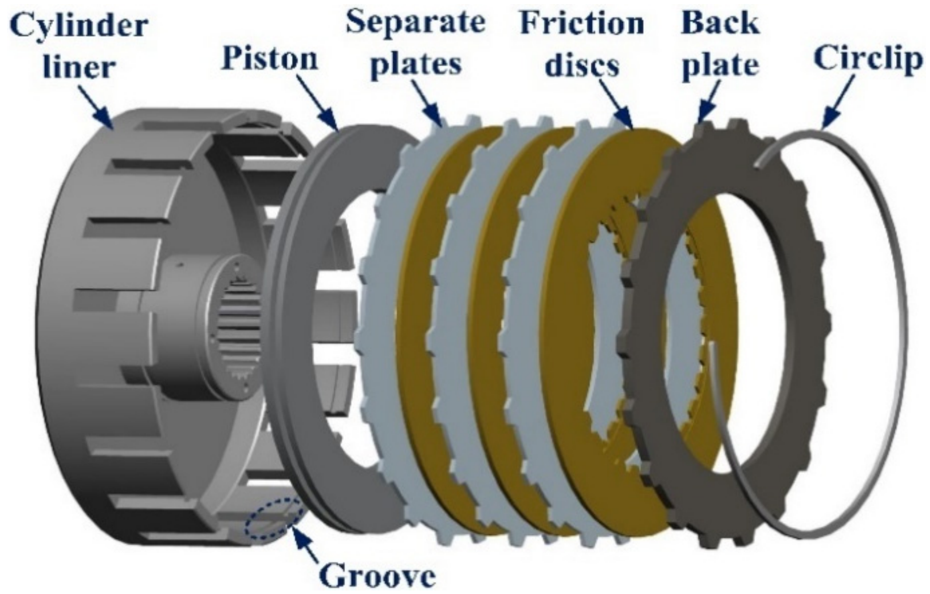


图2. 多盘离合器的分解布局图。

2.1. 花键摩擦

由于摩擦盘与分离板的结构特性相似，选择分离板进行理论分析与公式推导。如图3所示，在离合器匀速滑动过程中，分离板在摩擦力矩T_f与缸套阻力矩T_s的共同作用下达到平衡状态。作用于花键齿上的摩擦力、名义切向力与法向力分别表示为F_s、F_{cs}和F_{ns}。此外，F_p为施加于分离板的轴向接触压力。因此，平衡方程可写为

$$-T_s = T_f = -N_s F_{cs} R_s \tag{1}$$

其中R_s是节圆半径，N_s是单个分离片中的花键总数。应注意，顺时针方向为正，下标*s*和*f*分别代表分离板和摩擦片。

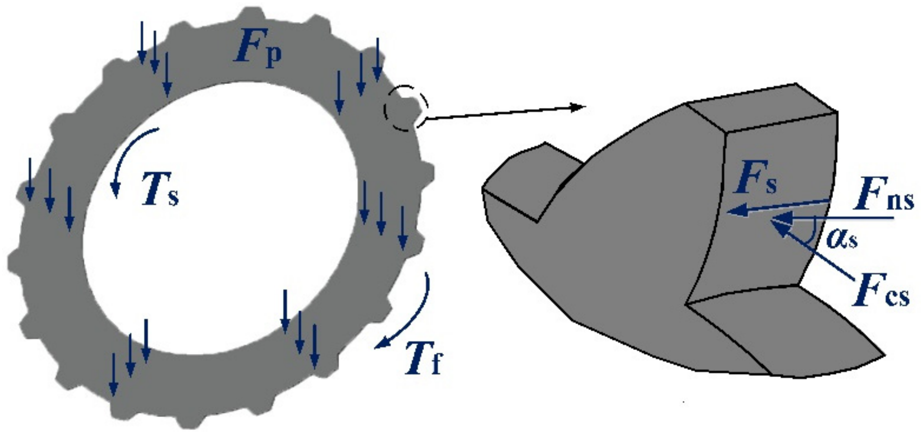


图3. 分离板的受力图。

缸套作用于渐开线花键的法向力可表示为

$$F_{ns} = \frac{F_{cs}}{\cos \alpha_s} \tag{2}$$

其中 α_s 是花键齿的压力角。

Therefore, the total friction force generated by spline teeth in a single separate plate can be expressed as

$$F_s = \mu_s N_s F_{ns} = \frac{\mu_s T_s}{R_s \cos \alpha_s} \tag{3}$$

where μ_s is the coefficient of friction (COF) of spline teeth.

2.2. Contact Pressure

As shown in Figure 4, the pressure p_c is evenly distributed on the piston. The friction components are numbered $0, 1, 2, \ldots, i, I+1, \ldots, M$, and the contact surfaces are similarly numbered $S_0, S_1, S_2, \ldots, S_M$. For example, the two surfaces of component 1 are represented as S_1, S_2, respectively. Due to the spline friction, the contact pressure acting on the friction components is smaller compared with the oil pressure p_c. To be more specific, the contact pressure gradually attenuates along the axial direction. Hence, the contact pressures on the two surfaces of friction component can be deduced as

$$p_{i+1} = p_i - \frac{F_s}{A} \tag{4}$$

where A is the contact area, namely, $A = \pi\left(R_2^2 - R_1^2\right)$; R_2 and R_1 are the outer and inner radii, respectively.

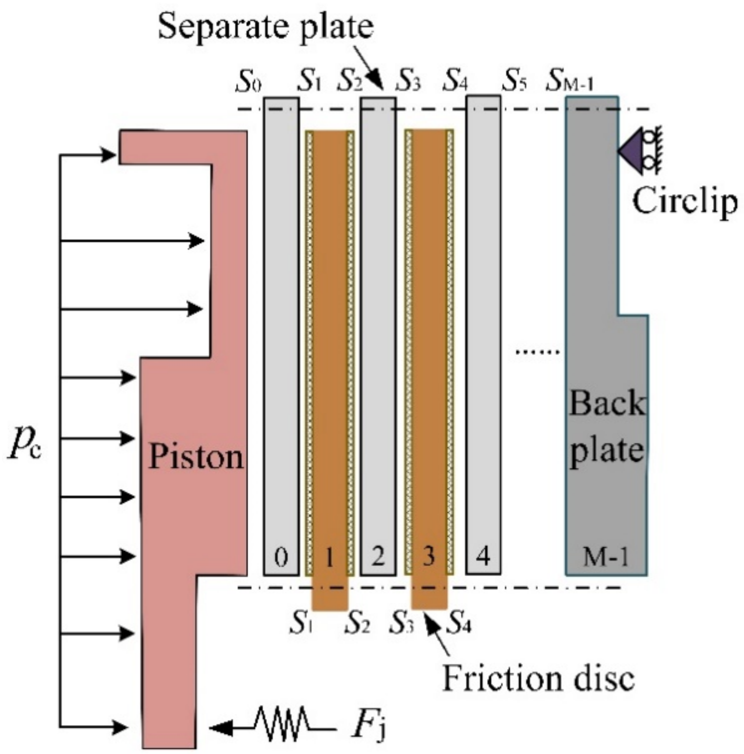


Figure 4. Simplified 2D model of the multidisc clutch.

2.3. Friction Torque

As the friction material of the clutch is the copper-based powder metallurgy material, the interface COF can be expressed as [18]:

$$\begin{aligned} \mu = {} & 23e^{\left(\frac{-2.6v}{(\ln\Theta-3.2)\left((28.3p)^{0.4}-0.87\right)}-5.16\right)} \\ & +0.08\left(e^{-0.005\Theta}-1\right)\left(e^{-0.2v}-1\right) \\ & +\frac{0.01\ln(4v+1)}{e^{0.005\Theta}} - 0.005\ln(28.3p) + 0.035 \end{aligned} \tag{5}$$

因此，单个分离片中花键齿产生的总摩擦力可表示为

$$F_s = \mu_s N_s F_{ns} = \frac{\mu_s T_s}{R_s \cos \alpha_s} \tag{3}$$

其中 μ_s 是花键齿的摩擦系数。

2.2. 接触压力

如图4所示，压力p_c 均匀分布在活塞上。摩擦元件编号为$0, 1, 2, \ldots, i, I+1, \ldots, M$，接触面类似地编号为$S_0, S_1, S_2, \ldots, S_M$。例如，元件1的两个面分别表示为$S_1$, S_2,。由于花键摩擦，作用在摩擦元件上的接触压力小于油压p_c。具体来说，接触压力沿轴向方向逐渐衰减。因此，摩擦元件两个面上的接触压力可推导为

$$p_{i+1} = p_i - \frac{F_s}{A} \tag{4}$$

其中A 是接触面积，即$A = \pi\left(R_2^2 - R_1^2\right)$；$R_2$ 和R_1 分别是外径和内径。

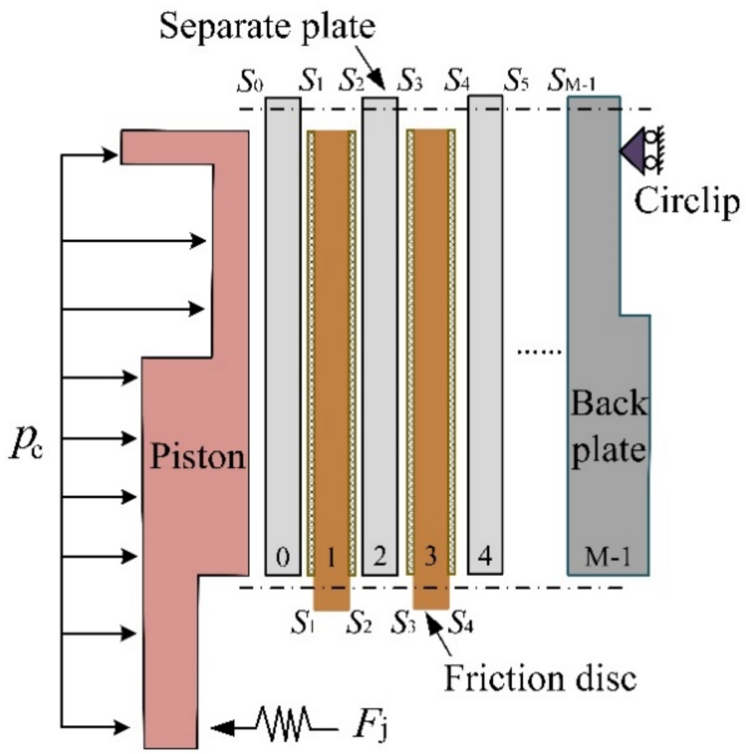


图4. 多盘离合器的简化二维模型。

2.3. 摩擦力矩

由于离合器的摩擦材料是铜基粉末冶金材料，界面摩擦系数可表示为 [18]:

$$\begin{aligned} \mu = {} & 23e^{\left(\frac{-2.6v}{(\ln\Theta-3.2)\left((28.3p)^{0.4}-0.87\right)}-5.16\right)} \\ & +0.08\left(e^{-0.005\Theta}-1\right)\left(e^{-0.2v}-1\right) \\ & +\frac{0.01\ln(4v+1)}{e^{0.005\Theta}} - 0.005\ln(28.3p) + 0.035 \end{aligned} \tag{5}$$

where ν, p and Θ are the relative sliding speed, average contact pressure and surface temperature, respectively.

Moreover, there is no relative motion between the piston, separate plate and back plate [19]. Hence, the FT on each friction component can be deduced as

$$T_i = \frac{2\pi\mu(R_2^3 - R_1^3)}{3} \times \begin{cases} p_{i+1}\,,\, i = 0 \\ (p_i + p_{i+1}),\, i = 1,2,3\ldots M-2 \\ p_i\,,\, i = M-1 \end{cases} \tag{6}$$

Integrating Equations (3)–(6), the contact pressure on the two adjacent surfaces can be expressed as

$$p_{i+1} = \begin{cases} \frac{1}{1+\xi} p_i\,,\, i = 0 \\ \frac{1-\xi}{1+\xi} p_i\,,\, i = 1,2,3\ldots M-2 \end{cases} \tag{7}$$

where ξ is the attenuation coefficient of contact pressure. Thus, the attenuation coefficient ξ_s of the separate plate can be written as

$$\xi_s = \frac{2\mu\mu_s(R_2^3 - R_1^3)}{3R_s(R_2^2 - R_1^2)\cos\alpha_s} \tag{8}$$

Without consideration of the spline friction, the contact pressure is evenly distributed and there is no axial pressure attenuation. Consequently, the total FT transmitted by the clutch can be expressed as [20]:

$$T_{none} = \frac{2\pi}{3}(M-1)\mu p_c(R_2^3 - R_1^3) \tag{9}$$

When the spline friction exists, the total FT transmitted by the clutch can be given as

$$T_{spline} = \frac{2\pi}{3}(R_2^3 - R_1^3)\sum_{i=1,3,5\ldots}^{M-1}\mu(p_i)\cdot p_i \tag{10}$$

3. Numerical Simulation

The physical parameters of the friction components are listed in Table 1. With the consideration of spline friction, the simulation is conducted to investigate the effect of applied pressure and the number of pairs on the interface COF and the axial attenuation characteristics.

Table 1. Physical parameters of the friction components.

Parameters	Nomenclature	Value
Inner-outer radii (m)	R_1, R_2	0.06, 0.073
Pitch radius (m)	R_f, R_s	0.057, 0.076
Pressure angle (°)	α_f, α_s	30, 27
Rotating speed (rpm)	n	2868
Linear speed of pitch radius (m/s)	v	18.2
Lubricating oil temperature (°C)	Θ	100

According to Equation (5), the variation of interface COF under different applied pressure is shown in Figure 5. The interface COF gradually decreases as the applied pressure varies from 0.2 MPa to 2 MPa. This phenomenon can be interpreted as follows. The increasing applied pressure leads to the rise of the asperity contact area between friction pairs, which makes the friction pair become relatively smooth [21]. Therefore, the interface COF decreases gradually.

其中 ν、p 和 Θ 分别是相对滑动速度、平均接触压力和表面温度。

此外，活塞、分离板和背板之间没有相对运动 [19]。因此，每个摩擦部件的摩擦力矩可推导为

$$T_i = \frac{2\pi\mu(R_2^3 - R_1^3)}{3} \times \begin{cases} p_{i+1}\,,\, i = 0 \\ (p_i + p_{i+1}),\, i = 1,2,3\ldots M-2 \\ p_i\,,\, i = M-1 \end{cases} \tag{6}$$

对方程(3)–(6)进行积分，两个相邻表面上的接触压力可表示为

$$p_{i+1} = \begin{cases} \frac{1}{1+\xi} p_i\,,\, i = 0 \\ \frac{1-\xi}{1+\xi} p_i\,,\, i = 1,2,3\ldots M-2 \end{cases} \tag{7}$$

其中 ξ 是接触压力的衰减系数。因此，分离板的衰减系数ξ_s 可写为

$$\xi_s = \frac{2\mu\mu_s(R_2^3 - R_1^3)}{3R_s(R_2^2 - R_1^2)\cos\alpha_s} \tag{8}$$

在不考虑花键摩擦的情况下，接触压力均匀分布且没有轴向压力衰减。因此，离合器传递的总摩擦力矩可表示为 [20]:

$$T_{none} = \frac{2\pi}{3}(M-1)\mu p_c(R_2^3 - R_1^3) \tag{9}$$

当存在花键摩擦时，离合器传递的总摩擦力矩可表示为

$$T_{spline} = \frac{2\pi}{3}(R_2^3 - R_1^3)\sum_{i=1,3,5\ldots}^{M-1}\mu(p_i)\cdot p_i \tag{10}$$

3. 数值模拟

摩擦组件的物理参数列于表1中。考虑花键摩擦，进行数值模拟以研究施加压力和摩擦副数对界面摩擦系数及轴向衰减特性的影响。

表1.摩擦组件的物理参数。

参数	术语表	值
内外半径 (m)	R_1, R_2	0.06, 0.073
节圆半径 (m)	R_f, R_s	0.057, 0.076
压力角 (°)	α_f, α_s	30, 27
转速 (rpm)	n	2868
节圆半径线速度 (m/s)	v	18.2
润滑油温度（°℃）	Θ	100

根据方程(5)，不同施加压力下界面摩擦系数的变化如图5所示。界面摩擦系数随着施加压力从0.2 MPa到2 MPa的变化逐渐减小。这一现象可以解释如下：施加压力的增加导致摩擦副间粗糙接触面积的增大，使摩擦副变得相对光滑 [21]。因此，界面摩擦系数逐渐减小。

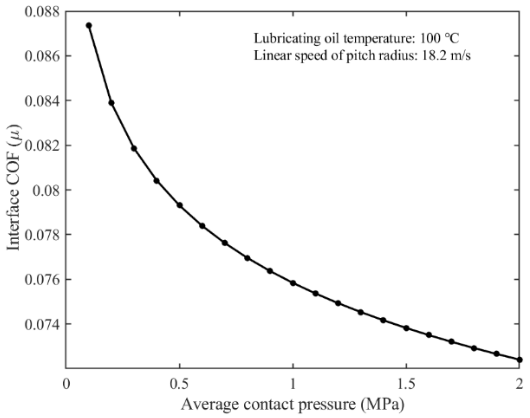


Figure 5. Variation of the interface coefficient of friction (COF) regarding the contact pressure.

According to Equations (5), (7) and (8), the interface COF and contact pressure under different working conditions are shown in Figures 6 and 7, respectively. As depicted in Figure 6, the farther the friction pair from the piston is, the greater the interface COF is. Moreover, as the applied pressure increases, the interface COF decreases gradually under the same number of friction pairs, which also has a good agreement with Figure 5.

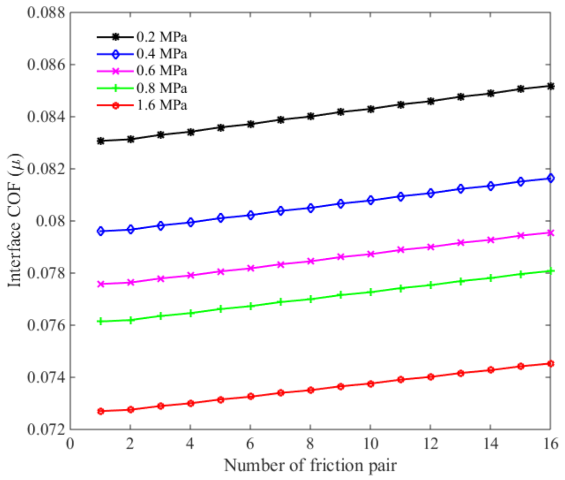


Figure 6. Variation of the interface COF regarding the friction pairs.

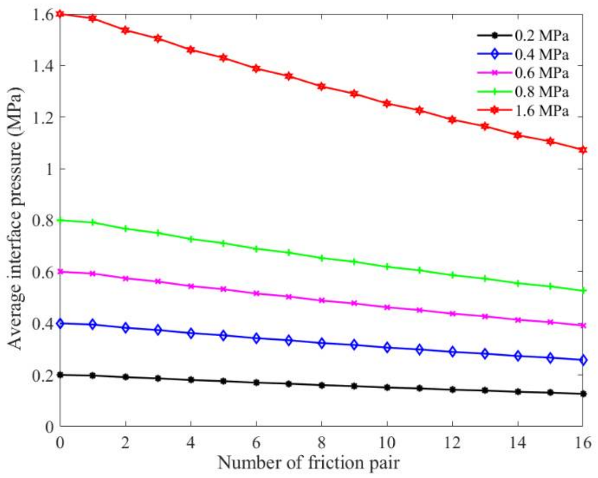


Figure 7. Attenuation of the contact pressure.

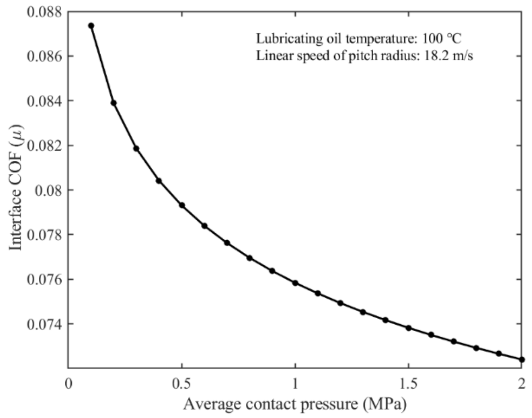


图5. 界面摩擦系数随接触压力的变化。

根据公式(5)、(7)和(8)，不同工况下的界面摩擦系数和接触压力分别如图6和7所示。如图6所示，摩擦副离活塞越远，界面摩擦系数越大。此外，随着施加压力的增加，在相同摩擦副数量下，界面摩擦系数逐渐减小，这与图5也吻合良好。

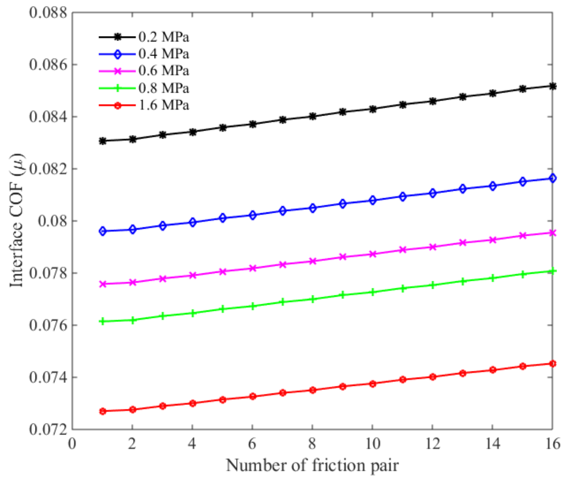


图6. 界面摩擦系数随摩擦 对。

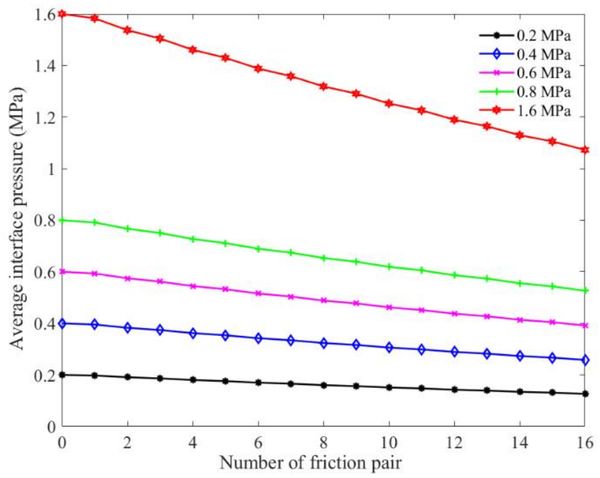


图7. 接触压力的衰减。

Moreover, the contact pressure decreases gradually along the axial direction with an increase in the number of the friction pairs, as shown in Figure 7. Additionally, the attenuation amplitude of contact pressure increases gradually with the applied pressure growing. For instance, the contact pressure decreases from 0.2 MPa to 0.131 MPa as the number of friction pair varies from 0 to 16, a reduction of 0.069 MPa. When the applied pressure is 1.6 MPa, the contact pressure acting on p_{16} decreases to 1.106 MPa, an attenuation of 0.494 MPa, indicating a significant effect of spline friction to the attenuation of contact pressure.

As shown in Figure 8, the FT increases linearly with the increasing number of friction pairs in the ideal case, and the more the friction pair is, the greater the FT is. Considering the spline friction, the growth rate of FT decreases gradually as the number of friction pairs increases. The reason is that the growth of interface COF is much smaller than the attenuation amplitude of contact pressure. Moreover, the greater the applied pressure is, the greater the loss of FT is. In other words, the spline friction will weaken the torque transmission capacity of multidisc clutch. For example, the reduction amplitude of total FT in a 16-friction-pair system are 18.5 N·m and 114.58 N·m under the applied pressure of 0.2 MPa and 1.6 MPa, respectively. Therefore, when the friction pair reaches a certain number, it is not reliable to realize a substantial enhancement of FT by continuing to add the friction components.

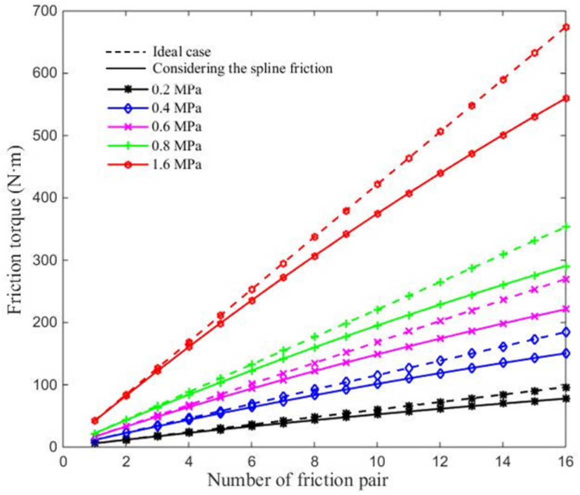


Figure 8. Variation of the friction torque.

4. Attenuation Experiment

4.1. Experimental Equipment

As depicted in Figure 9, the test bench is mainly composed of the motor, the clutch pack, the hydraulic system and the data collection and control system. The pressure exerted on the piston is applied by the air pump. The lubricating oil flows into the clutch to lubricate and cool the friction components, and the heat exchanger is used to maintain the oil temperature at 100 °C. In addition, the data collection and control system plays a role in monitoring the real-time signals, including the rotating speed, the FT, the applied pressure and temperature, etc. Moreover, it can also calculate the equivalent COF μ^* of friction pair according to the total FT T^* measured by the torque sensor, the equivalent COF can be given as

$$\mu^* = \frac{3T^*}{2\pi p_c(R_2^3 - R_1^3)(M-1)} \tag{11}$$

此外，接触压力沿轴向方向随摩擦副数量的增加而逐渐减小，如图7所示。另外，接触压力的衰减幅度随施加压力的增大而逐渐增加。例如，当摩擦副数量从0变化到16时，接触压力从0.2 MPa降至0.131 MPa，减少了0.069 MPa。当施加压力为1.6 MPa时，作用于 p_{16} 的接触压力降至1.106 MPa，衰减了0.494 MPa，表明花键摩擦对接触压力的衰减有显著影响。

如图8所示，在理想情况下，摩擦力矩随摩擦副数量的增加而线性增加，摩擦副越多，摩擦力矩越大。考虑花键摩擦时，摩擦力矩的增长率随摩擦副数量的增加而逐渐减小。原因是界面摩擦系数的增长远小于接触压力的衰减幅度。此外，施加压力越大，摩擦力矩的损失越大。换句话说，花键摩擦会削弱多盘离合器的扭矩传递能力。例如，在施加压力为0.2 MPa和1.6 MPa时，16摩擦副系统中的总摩擦力矩下降幅度分别为18.5 N·m和114.58 N·m。因此，当摩擦副达到一定数量时，通过继续增加摩擦元件来大幅提升摩擦力矩并不可靠。

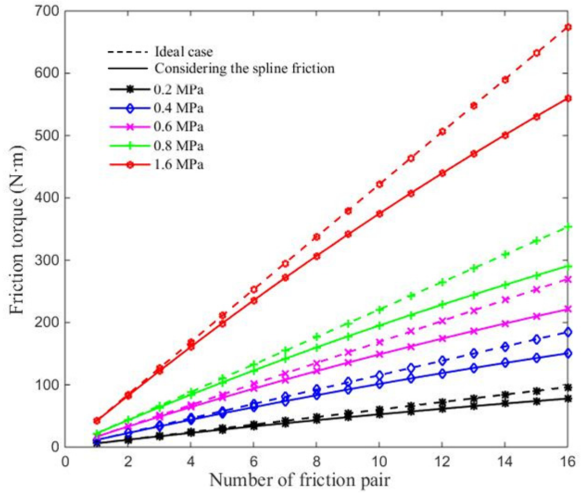
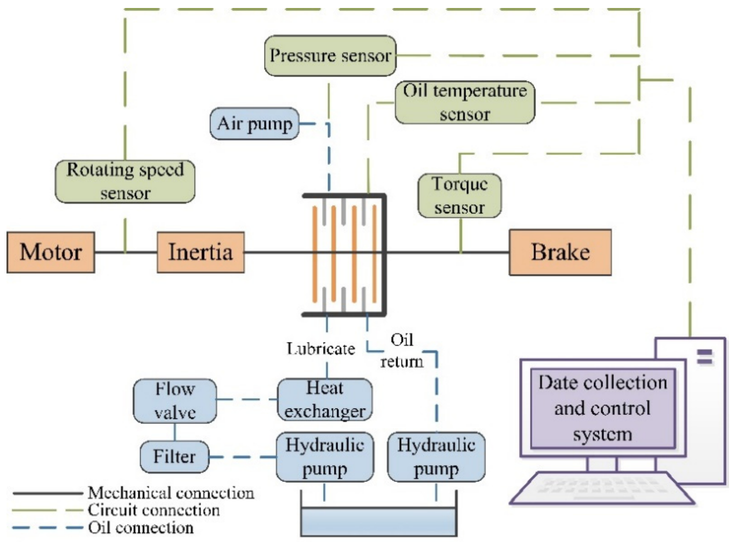


图8. 摩擦力矩的变化。

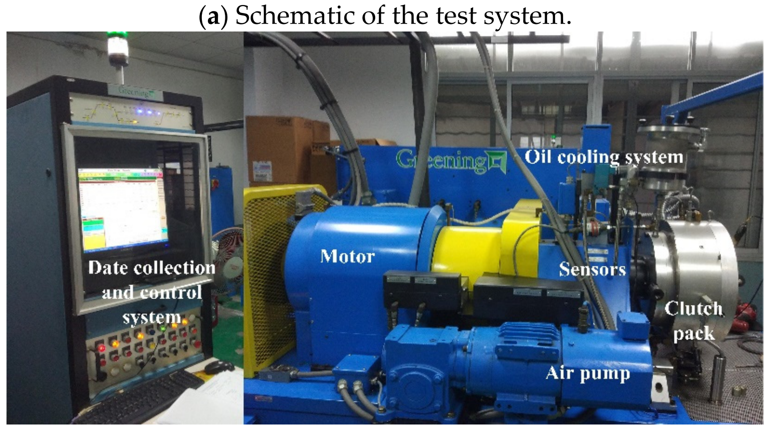
4. 衰减实验*4.1.* 实验设备

如图 9所示，试验台主要由电动机、离合器组件、液压系统以及数据采集与控制系统组成。活塞上施加的压力由气泵提供。润滑油流入离合器以润滑和冷却摩擦组件，换热器用于将油温维持在 100 °℃。此外，数据采集与控制系统负责监测实时信号，包括转速、摩擦力矩、施加压力和温度等。同时，它还能根据扭矩传感器测得的总摩擦力矩 μ^* 计算摩擦副的等效摩擦系数 T^* ，等效摩擦系数可表示为

$$\mu^* = \frac{3T^*}{2\pi p_c(R_2^3 - R_1^3)(M-1)} \tag{11}$$



(**a**) Schematic of the test system.



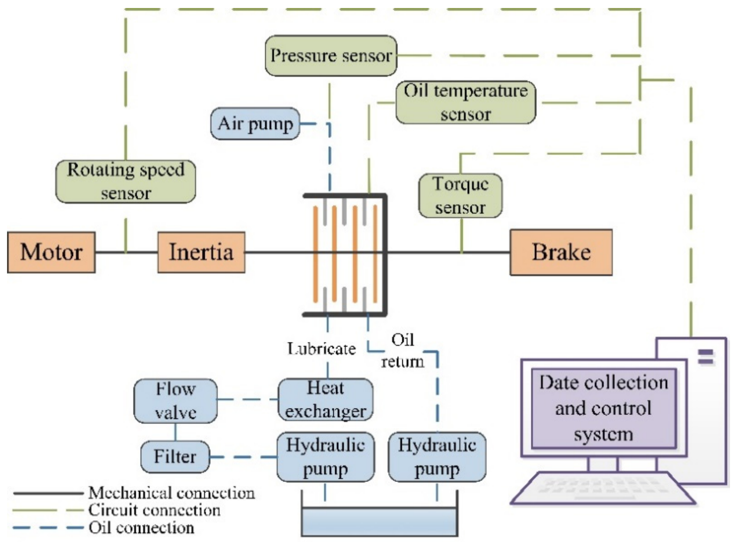
(**b**) Test bench of the clutch.

Figure 9. Test system of the multidisc clutch.

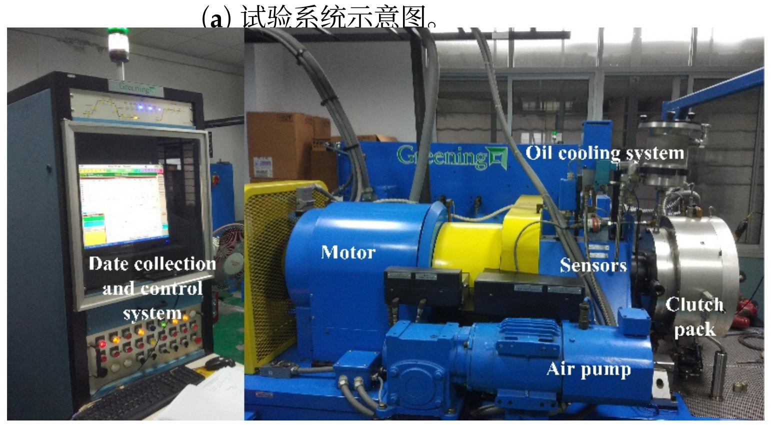
The separate plate is made of the 65 Mn steel, while the friction discs are made of copper-based powder metallurgy material with nano-modifier, and the physical parameters are shown in Table 1. Moreover, the type of the lubricating oil is 10W/40-CF, while the average unit flow rate is 4 mL/(min·cm^2).

4.2. Experimental Method

Before the formal experiment, a running-in test is essential to ensure the equipment works properly to collect the valid data. The arrangement of clutch pack is divided into two categories, namely, the two-friction pair system and six-friction pair system. During the experiment of the former system, the initial motor speed is 2686 rpm. Moreover, an experiment cycle consists of five engagement tests at the applied pressure conditions of 0.2 MPa, 0.4 MPa, 0.6 MPa, 0.8 MPa, and 1.6 MPa, respectively. In order to reduce the measurement error, a total of 100 experiment cycles are carried out successively. Subsequently, the same experiment scheme is also conducted in the six-friction pair system. Figure 10 shows the experiment conditions.



(**a**) 试验系统示意图。



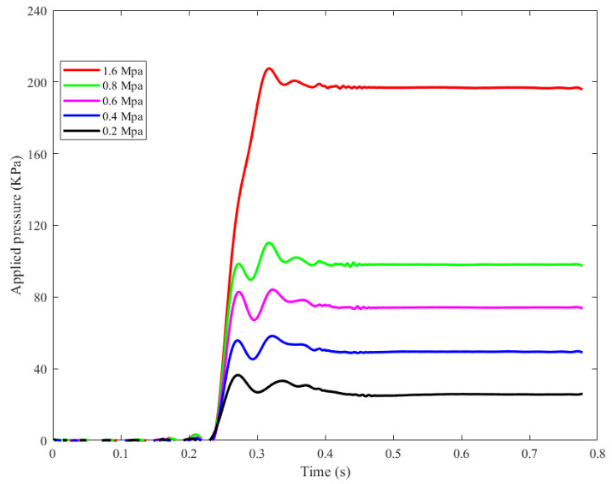
(**b**) 离合器试验台。

图9. 多盘离合器的测试系统。

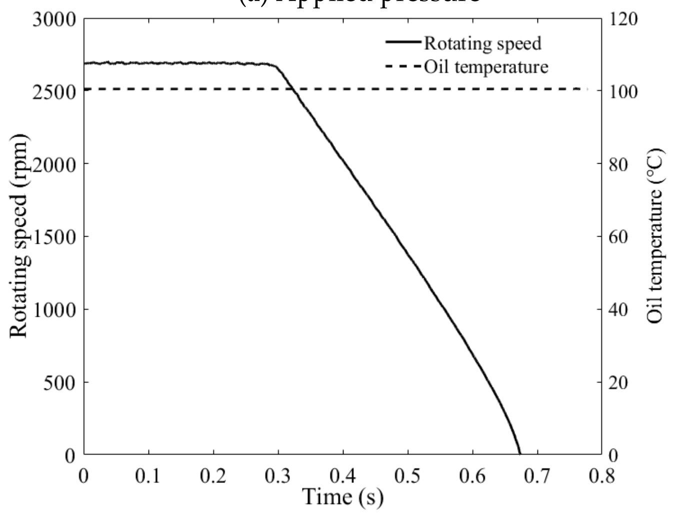
分离板由65Mn钢制成，而摩擦盘由含纳米改性剂的铜基粉末冶金材料制成，物理参数见表1。此外，润滑油型号为10W/40-CF，平均单位流量为4 mL/(min·cm^2)。

4.2. 实验方法

在正式实验之前，必须进行磨合试验以确保设备正常工作，从而收集有效数据。离合器组件的布置分为两类，即双摩擦副系统和六摩擦副系统。在前者系统的实验中，初始电机转速为2686 rpm。此外，一个实验周期包括在施加压力条件分别为0.2 MPa、0.4 MPa、0.6 MPa、0.8 MPa和1.6 MPa下的五次接合试验。为了减少测量误差，总共连续进行100个实验周期。随后，在六摩擦副系统中也执行相同的实验方案。图10显示了实验条件。



(**a**) Applied pressure

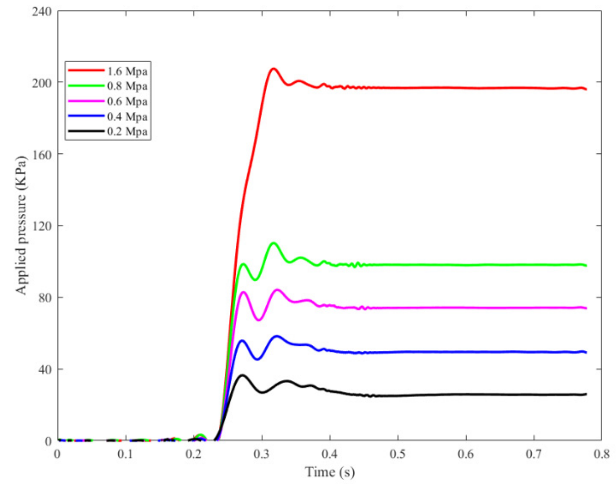


(**b**) Rotating speed and lubricating temperature

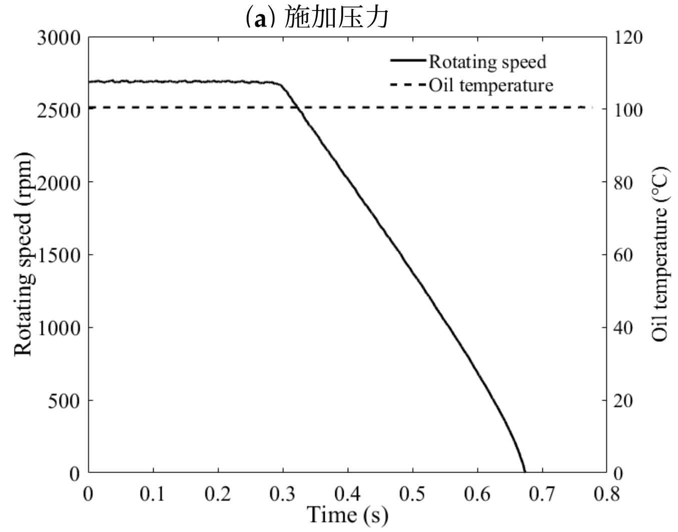
Figure 10. Experiment conditions.

5. Results and Discussion

After experiments, the results of the equivalent COF and FT in the two-friction pair and six-friction pair system are shown in Figures 11 and 12, respectively. The mean values of the equivalent COF, friction torques and single-pair friction torques in that two systems are shown in Table 2.



(**a**) 施加压力

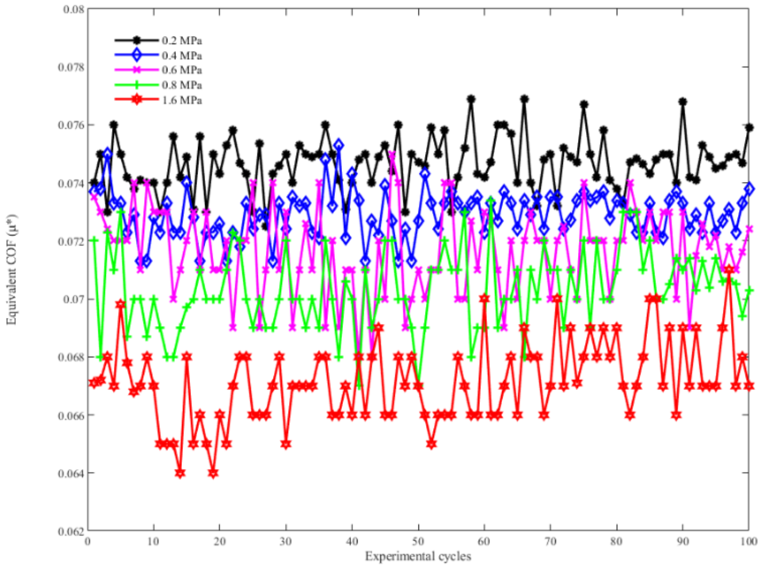


(**b**) 转速和润滑温度

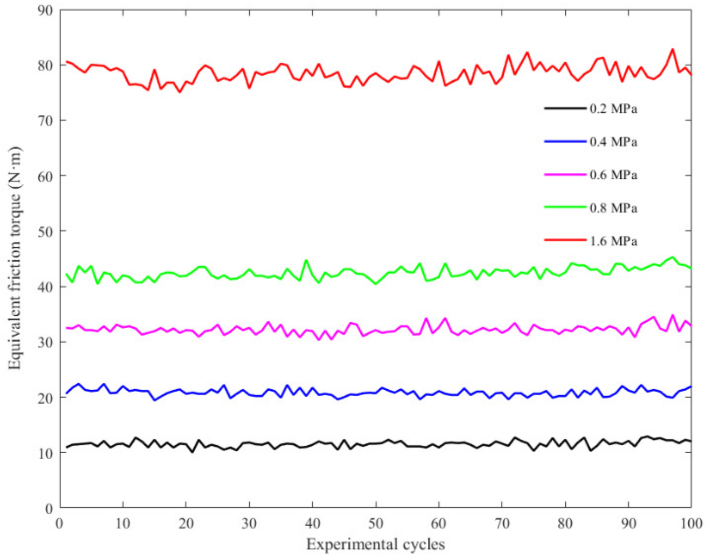
图10. 实验条件。

5. 结果与讨论

实验后，两摩擦副和六摩擦副系统中的等效摩擦系数和摩擦力矩结果分别如图11 和12所示。两个系统中等效摩擦系数、摩擦扭矩和单对摩擦扭矩的平均值见表2。

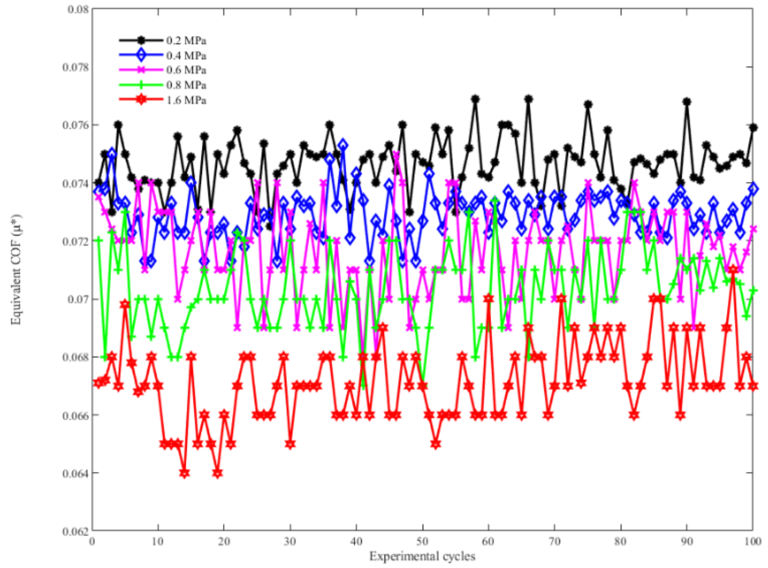


(**a**) Equivalent COF

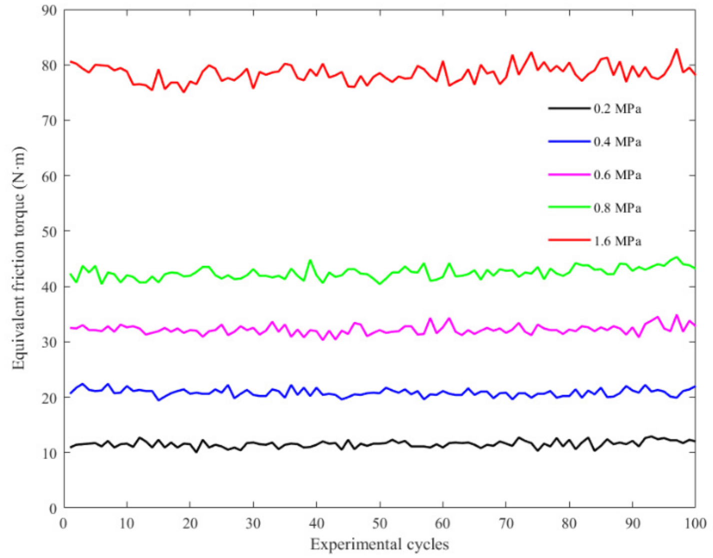


(**b**) Equivalent friction torque

Figure 11. Two-friction pair system.

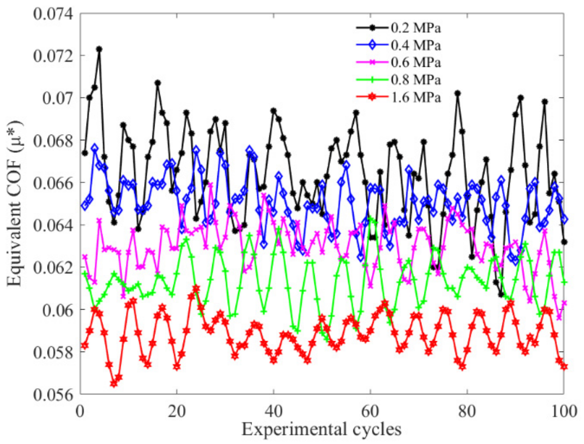


(**a**) 等效摩擦系数

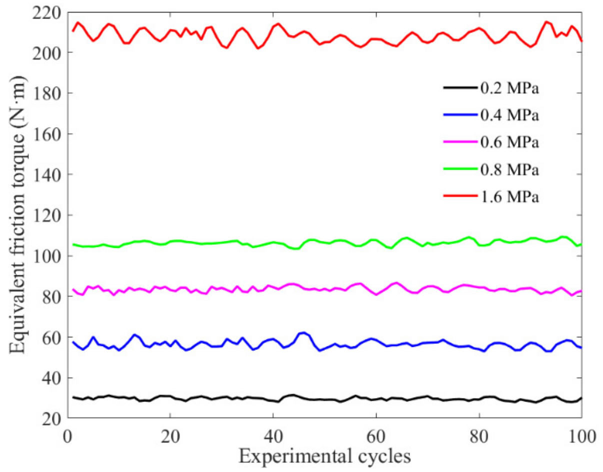


(**b**) 等效摩擦力矩

图11. 双摩擦副系统。



(**a**) Equivalent COF.



(**b**) Equivalent friction torque.

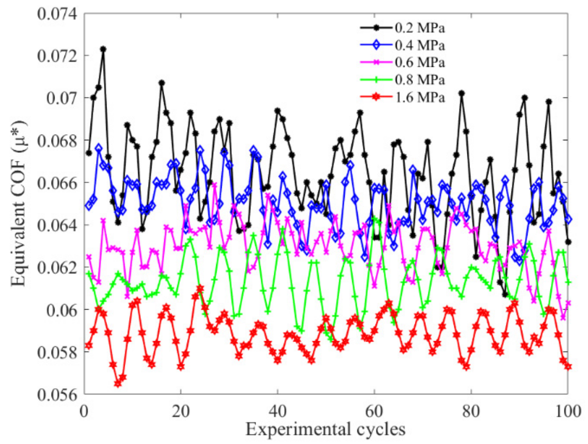
Figure 12. Six-friction pair system.

Table 2. Mean values of the two systems.

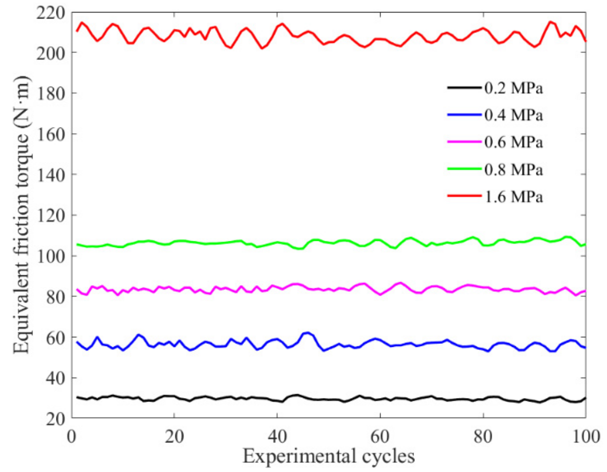
Applied Pressure (MPa)		**0.2**	**0.4**	**0.6**	**0.8**	**1.6**
Two-friction pair system	μ_2*	0.07444	0.07271	0.07143	0.07091	0.06696
	T_2* (N·m)	11.025	21.648	31.914	42.247	79.914
	$\overline{T}_2$ (N·m)	5.5123	10.824	15.957	21.123	39.957
Six-friction pair system	μ_6*	0.06685	0.06545	0.06315	0.06093	0.05857
	T_6* (N·m)	29.394	56.258	83.287	106.45	207.35
	$\overline{T}_2$ (N·m)	4.899	9.376	13.881	17.742	34.558

As the applied pressure increases, the equivalent COF gradually decreases, while the total FT increases nonlinearly. The equivalent COF in the two-friction pair system is greater than the other one under the same applied pressure. Moreover, the total FT in six-friction pair system does not reach three times as much as that in the two-friction pair system, indicating that the increase of friction pairs will enlarge the attenuation amplitude of axial contact pressure. In addition, the increasing applied pressure brings about the gradual decrease of the growth rate of average single-pair FT. The average single-pair FT $\overline{T}_2$ is large than $\overline{T}_6$ under various applied pressure conditions due to the spline friction. Therefore, the total FT cannot be significantly improved by increasing the applied pressure or using more friction pairs.

According to Figures 6 and 7, the contact pressure and interface COF of the second friction pair is close to the first one, respectively. For example, when the applied pres-



(**a**) 等效摩擦系数。



(**b**) 等效摩擦力矩。

图12. 六摩擦副系统。

表2. 两个系统的平均值。

施加压力(MPa) 0.2 0.4 0.6 0.8 1.6	
双摩擦副系统	μ_2* 0.07444 0.07271 0.07143 0.07091 0.06696 T_2*(N··米) 11.025 21.648 31.914 42.247 79.914 $\overline{T}_2$(牛·米) 5.5123 10.824 15.957 21.123 39.957
六摩擦副系统	μ_6* 0.06685 0.06545 0.06315 0.06093 0.05857 T_6*(N·m) 29.394 56.258 83.287 106.45 207.35 T_2 (N·m) 4.899 9.376 13.881 17.742 34.558

随着施加压力增大，等效摩擦系数逐渐减小，而总摩擦力矩呈非线性增加。在相同施加压力下，双摩擦副系统中的等效摩擦系数大于另一个系统。此外，六摩擦副系统中的总摩擦力矩并未达到双摩擦副系统的三倍，表明摩擦副数量的增加会放大轴向接触压力的衰减幅度。另外，施加压力的增大会导致平均单对摩擦扭矩的增长率逐渐下降。由于花键摩擦，在各种施加压力条件下，平均单对摩擦扭矩T_2 都大于T_6 。因此，通过增加施加压力或使用更多摩擦副并不能显著提高总摩擦力矩。

根据图6 和7，第二摩擦副的接触压力和界面摩擦系数分别接近第一摩擦副。例如，当施加压力为

sure is 0.2 MPa, the ratios of the first two contact pressure and interface COF are 96.7% and 100.24%, severally. Therefore, the average equivalent COF in the two-friction pair system can be regarded as the interface COF of the first pair in the six-friction pair system. Besides, the range of spline COF μ_{fs} is 0.1–0.15 when the spline of friction component is in static friction status [20]. Accordingly, the total calculated torque in six-friction pair system and the relative error between the theoretical and experimental value can be obtained as shown in Table 3.

Table 3. Total calculated torque and relative error.

Applied Pressure (MPa)		0.2	0.4	0.6	0.8	1.6
Calculated torque (N·m)	$\mu_{fs} = 0.1$	30.747	60.136	88.694	117.44	222.4
	$\mu_{fs} = 0.15$	29.977	58.664	86.56	114.64	217.37
Relative error (%)	$\mu_{fs} = 0.1$	4.6	6.89	6.49	10.3	7.26
	$\mu_{fs} = 0.15$	1.98	4.28	3.93	7.69	4.83

Under the same applied pressure, the calculated torque decreases gradually with the increasing value of static COF. Besides, as the applied pressure increases, the total calculated torque increases approximately linearly. Moreover, the maximum relative error is within 10.3%, which not only verifies the reliability of theoretical model, but also testifies the significant effect of spline friction on the torque transmission capacity.

6. Conclusions

Considering the spline friction of friction components, the detailed numerical model and experimental method were established to study the FT attenuation in a wet multi-disc clutch. Under various working conditions, the contact pressure and FT obtained by bench test show a good agreement with the numerical model, which also verifies the existence of spline friction. The conclusions can be summarized as follows: the spline friction is an important factor affecting the attenuation of the axial contact pressure and FT of the friction components. With the increase of applied pressure, the attenuation of the contact pressure along the axial direction increases gradually, whereas the growth rate of the average single-pair FT decreases. In addition, the average single-pair FT decreases gradually with the increasing friction pairs under the same applied pressure. To be more specific, it is not reliable to obtain a substantial enhancement of FT by continuing to add friction components when the friction pair reaches a certain number. Therefore, in order to reduce the effect of spline friction, it suggests to choose the reasonable structural parameters for friction component, to improve the manufacturing processes of spline and to increase the lubrication flow at the spline.

Author Contributions: M.C. is responsible for the theoretical modeling and research funding. B.Z. did the academic writing and the numerical analysis. Y.F. not only formulated the specific experimental scheme, but also provided the academic and experimental guidance. L.W. and H.W. did part of the experiments and organized the experimental data. All authors have read and agreed to the published version of the manuscript.

Funding: This work was supported by the National Natural Science Foundation of China (Grant NO. 51775045 and NO. 51975047), and the basic product innovation research project of the Ministry of Industry and Information Technology of China (JCCPCX201705).

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: Not applicable.

Conflicts of Interest: The authors declare no conflict of interest.

0.2 MPa时，前两个接触压力和界面摩擦系数的比值分别为96.7%和100.24%。因此，双摩擦副系统中的平均等效摩擦系数可视为六摩擦副系统中第一对的界面摩擦系数。此外，当摩擦元件的花键处于静摩擦状态时，花键摩擦系数 μ_{fs} 的范围为0.1–0.15 [20]。据此，可得到六摩擦副系统中的总计算扭矩以及理论值与实验值之间的相对误差，如表3所示。

表3. 总计算扭矩和相对误差。

施加压力 (MPa)		0.2	0.4	0.6	0.8	1.6
计算出的 (N·m)	$\mu_{fs} = 0.1$	30.747	60.136	88.694	117.44	222.4
	$\mu_{fs} = 0.15$	29.977	58.664	86.56	114.64	217.37
相对误差 (%)	$\mu_{fs} = 0.1$	4.6	6.89	6.49	10.3	7.26
	$\mu_{fs} = 0.15$	1.98	4.28	3.93	7.69	4.83

在相同施加压力下，计算扭矩随静摩擦系数增大而逐渐减小。此外，随着施加压力增加，总计算扭矩近似线性增加。而且最大相对误差在10.3%以内，不仅验证了理论模型的可靠性，还证明了花键摩擦对扭矩传递能力的显著影响。

6. 结论

考虑摩擦组件的花键摩擦，建立了详细的数值模型和实验方法，以研究湿式多片离合器中的摩擦力矩衰减。在不同工况下，通过台架试验获得的接触压力和摩擦力矩与数值模型吻合良好，这也验证了花键摩擦的存在。结论可总结如下：花键摩擦是影响摩擦组件轴向接触压力和摩擦力矩衰减的重要因素。随着施加压力的增加，接触压力沿轴向方向的衰减逐渐增大，而平均单对摩擦扭矩的增长率则减小。此外，在相同施加压力下，平均单对摩擦扭矩随着摩擦副数量的增加而逐渐减小。更具体地说，当摩擦副达到一定数量时，继续增加摩擦组件并不会显著提升摩擦力矩。因此，为减少花键摩擦的影响，建议选择合理的摩擦组件结构参数，改进花键的制造工艺，并增加花键处的润滑流量。

作者贡献：M.C.负责理论值建模和研究经费。B.Z.完成学术写作和数值分析。Y.F.不仅制定了具体的实验方案，还提供了学术和实验指导。L.W.和H.W.进行了部分实验并整理了实验数据。所有作者均已阅读并同意稿件发布版本。

资金：本研究得到国家自然科学基金（批准号：51775045和51975047）以及中华人民共和国工业和信息化部基础产品创新研究项目（JCCPCX201705）的支持。

机构审查委员会声明：不适用。

知情同意声明：不适用。

数据可用性声明：不适用。

利益冲突：作者声明没有利益冲突。

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